

A Research Article Template for Mathematical Writing

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Abstract

This template provides a structured framework for preparing mathematical research articles using $\text{\LaTeX}$. It demonstrates the organization of mathematical results, theorem environments, equations, tables, figures, cross-references, and bibliographic references.

Keywords: LaTeX, mathematical writing, research article, mathematical analysis, academic publishing

1 Introduction

$\text{\LaTeX}$ is widely used for preparing mathematical and scientific manuscripts because it provides precise control over mathematical notation and document structure.

A research article can be organized into clearly defined sections containing definitions, lemmas, propositions, theorems, proofs, examples, and conclusions.

This template is designed to demonstrate such a workflow in a reusable format.

2 Preliminaries

We begin with a standard definition.

Definition 2.1. Let I be an interval in $\mathbb{R}$. A function $f : I \rightarrow \mathbb{R}$ is said to be convex if

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y) \tag{1}$$

for all $x, y \in I$ and $t \in [0, 1]$.

Equation (1) illustrates how mathematical expressions can be labelled and referenced automatically.

3 Main Results

The following theorem illustrates the presentation of a mathematical result.

Theorem 3.1. *Let $f : [a, b] \rightarrow \mathbb{R}$ be a convex function. Then, for every $x, y \in [a, b]$ and $t \in [0, 1]$,*

$$f(tx + (1 - t)y) \leq tf(x) + (1 - t)f(y). \quad (2)$$

Proof. The result follows directly from the definition of convexity given in Definition 2.1. $\square$

4 Supporting Results

Lemma 4.1. *Let $x, y \in \mathbb{R}$. Then*

$$(x + y)^2 = x^2 + 2xy + y^2. \quad (3)$$

Proof. Expanding the product gives

$$(x + y)^2 = (x + y)(x + y) \quad (4)$$

$$= x^2 + xy + yx + y^2 \quad (5)$$

$$= x^2 + 2xy + y^2. \quad (6)$$

$\square$

5 Examples

Example 5.1. Consider the function

$$f(x) = x^2. \quad (7)$$

Since

$$f''(x) = 2 > 0, \quad (8)$$

the function is convex on $\mathbb{R}$.

6 Illustrative Table

Table 1 demonstrates a simple research-style table.

Table 1: Illustrative mathematical quantities

Quantity	Symbol	Value
Length	L	10
Width	W	5
Area	A	50

7 Figures

Figures can be incorporated into a manuscript using the `graphicx` package.

A typical figure can be inserted as follows:

```
\begin{figure}[htbp]
\centering
\includegraphics[width=0.75\textwidth]{figure-name}
\caption{Illustrative figure.}
\label{fig:illustrative}
\end{figure}
```

The figure can then be referenced using Figure ??.

8 Cross-Referencing

Cross-referencing is particularly useful in mathematical manuscripts.

For example:

- Definition 2.1 introduces convexity.
- Theorem 3.1 presents a mathematical result.
- Lemma 4.1 provides a supporting identity.
- Table 1 demonstrates table formatting.

The `cleveref` package can also be used for convenient automatic referencing of mathematical objects.

9 Conclusion

This template demonstrates a structured workflow for preparing mathematical research manuscripts using $\text{\LaTeX}$.

The template includes:

- academic document structure;
- mathematical notation;
- definitions and theorem environments;
- proofs and supporting results;
- numbered equations;
- tables and figures;
- labels and cross-references;
- bibliography integration;
- reusable document organization.